

LISTENING

#27

1 – Listen to a science reporter explaining an unusual event. Pay attention to what normally happens during a supernova, what made this one different, and what scientists believe might have caused the strange behavior.

The Most Unusual Supernova

1 - What does the science reporter say normally happens when a star reaches the end of its life?

- a) It becomes a planet
- b) It either collapses into a black hole or explodes as a supernova
- c) It disappears without any energy release

2 - How much energy does a typical supernova release?

3 - How long does a normal supernova last?

4 - What was unusual about this supernova's duration?

- a) It wasn't bright at all
- b) It continued for 600 days
- c) It exploded twice

5 - What does the reporter say the star may have produced in its core?

- a) Ice
- b) Antimatter
- c) Gas clouds

6 - According to the audio, what happens when antimatter is produced inside a star?

7 - What does the reporter say usually remains after one massive supernova?

- a) A small neutron star
- b) A new galaxy
- c) Nothing remains

8 - How long might this unusual star have been releasing energy in total?

- a) 60 years or longer
- b) Less than one day
- c) Exactly 600 days

9 - Why is it difficult for scientists to study stars like this one?

Full Name _____

READING COMPREHENSION

Newborn supernova spotted only hours after its explosive birth

Supernovae are among the most dramatic events in the universe. When massive stars run out of fuel, they can collapse or explode, producing powerful bursts of energy and light. These explosions often outshine entire galaxies and play a key role in creating heavy elements — the building blocks of planets and even life.

Yet, not all supernovae behave in the “normal” way. Sometimes astronomers observe events that last much longer, behave more erratically, or shine brighter than expected. These strange cases challenge existing ideas about how stars live — and die.

For example, recent observations suggest that a massive star might explode not because it suddenly collapses, but because of interactions with a companion object — perhaps a black hole. In such a “binary system,” the black hole’s gravity can distort the star, stripping away its outer layers or even stretching it before it explodes. This may result in a supernova that’s slower, more chaotic, or unusual compared to standard models.

In other cases, the material that the star ejected before the explosion can collide with the blast wave. When ejecta hits this dense circumstellar material, it can produce extra light and make the supernova last longer or become more luminous. Such interactions force scientists to rethink the traditional single-burst view of stellar death.

Studying these peculiar supernovae helps astronomers refine theories of stellar evolution — how stars are born, live, and eventually die. They remind us that nature is often more imaginative than textbooks.

Retrieved from: <https://www.popsci.com/science/newborn-supernova>

Comprehension Questions

1. Why are supernovae important for the universe?

2. What makes some supernovae “unusual,” according to the text?

3. What might cause a star to explode differently, rather than following the usual pattern?

4. What can happen when a star’s ejected material collides with a supernova blast wave?

5. Do you think scientists should spend more time observing unusual supernovae rather than just the “normal” ones? Why or why not?
